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Disturbance rejection control of shaking table based on adaptive-model-based state estimator[☆]

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ABSTRACT

Real-time hybrid simulation (RTHS) is an important method for the seismic testing of building structures. This method requires high control accuracy for the waveform reproduction of a shaking table. An active disturbance rejection control (ADRC) method is introduced to control the shaking table in this research. In view of the difficulty of parameters setting caused by the large deviation of the extended state observer (ESO) to the state estimation of the shaking table, a model-based adaptive state estimator (AMSE) is designed that uses the model information to estimate the state, and introduces an adaptive strategy to dynamically compensate for the amplitude and phase deviation of the model. The total disturbance is observed by the ESO. Additionally, an AMSE-based ADRC is proposed to achieve high-precision seismic waveform reproduction. The separation principle of the designed system is proved by the Routh–Hurwitz criterion, and the pole assignment bandwidth method is proposed to improve the control performance of the system by flexibly assigning the controller poles. Finally, the shaking table experiments are designed for different specimens. These experiments verify the advantages of the proposed method for state estimation and parameter setting and can effectively improve the stability and control accuracy of shaking table control.

1. Introduction

An earthquake is a type of sudden disaster with great destructiveness, high prevention difficulty, wide distribution and serious threat to life and property. Seismic structure tests are very important for understanding the performance, reaction process, and failure mechanism of building structures during earthquakes. Full-scale structural tests are very difficult to carry out because they are expensive and limited by the capacity of the testing equipment. Real-time hybrid simulation (RTHS) is a cyber–physical testing technique for investigating especially large or complicated structural systems, and it has been used to study the seismic performance of large structures (Zhou et al., 2019).

The studied objects in RTHS are divided into an experimental substructure and a numerical substructure (Li, Zhou, & Li, 2021). The experimental substructure includes a loading device and a part of the specimen that cannot be simulated exactly and needs to be experimentally tested. The numerical substructure, which is realized by numerical model, mainly includes the part of the specimen that does not require a performance test. In the RTHS experiment, through the exchange of displacement and force sensors and simulation calculation information, the physical and digital subsystems interact in real time to complete the performance test of the specimen (Li et al., 2022; Wang, Ning, et al., 2019).

In the RTHS of an aseismic structure, the loading of the experimental substructure is usually achieved by a shaking table. The waveform reproduction accuracy of the shaking table plays a decisive role in the effectiveness and stability of the RTHS (Xu et al., 2019). In experiments for different specimens, the order and characteristics of the specimens and the force on the shaking table will change (Chen et al., 2022). The variety of specimens in the RTHS requires the control algorithm of the shaking table to have strong adaptability to different specimens and working conditions (Zhou et al., 2023).

Active disturbance rejection control (ADRC) is a practical algorithm with minimal dependence on the model information, low sensitivity to model parameters, and good robustness and disturbance rejection capabilities (Gao, 2006). The core idea of ADRC is to estimate the total disturbance of the system through an extended state observer (ESO),

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which is a component of the ADRC, and eliminate it with feedback control (Sun & You, 2016). In order to avoid complex parameter tuning, linear ADRC (LADRC) is generally used in practical engineering applications. LADRC adjusts the parameters with the bandwidth method, simplifies parameter tuning, and greatly promotes the application of ADRC (Gao, 2013).

For LADRC, studies show that the ESO has an inherent overshoot phenomenon when estimating higher-order states. The higher the order is, the larger the overshoot is, and the lower the observation accuracy of the state becomes (Shao & Wang, 2015). The presence of field noise also limits the bandwidth of the ESO. These factors result in tuning difficulty of the control law parameters using the ESO observation states. The bandwidth method used by LADRC assigns the controller poles to the same pole of the negative real axis, which simplifies the parameter tuning but also limits the control performance of the ADRC (Wu et al., 2022). To solve the above problems, researchers have carried out a considerable amount of research on the improvement of parameter setting methods and the performance of ESO estimation (Liu et al., 2022; Rubén & Luis, 2021).

In order to improve the performance of the ESO, Xue et al. (2015) combined the adaptive method with the ESO and adjusted the parameters of ESO online with the adaptive algorithm to improve the accuracy of state and total disturbance estimation. However, there are some difficulties in determining the boundary conditions of adaptive parameters. Wei, Zhang, and Zuo (2021) introduced the phase leading compensation into the total disturbance output of the ESO to make the disturbance estimation more accurate. Fu et al. added the information for the model to the estimation of the extended state when designing the ESO (Fu & Tan, 2016). Sun et al. (2016) proposed a model-assisted ADRC method for non-minimum phase systems and obtained good results through simulation. The difficulty of control system design can be effectively reduced by using the known model information (Li, Zhu, Mao, et al., 2021; Plummer, 2016), but it is often difficult for this method to achieve the expected effect when the sensors of the experiment are noise-polluted. Shi et al. (2022) proposed an extended state filter that solved the high-frequency noise problem. However, its simulation verification of high-order state observation was only verified to the second order. These improvements were limited to total disturbance, and the estimation of each order state was neglected. The inherent overshoot mentioned in the literature (Shao & Wang, 2015) still exists, and it seriously affects the performance of feedback control.

In order to make full use of the model information and avoid noise pollution, some scholars have tried to apply the known model information to vibration table state estimation (Carrion & Spencer, 2006; Verma & Sivaselvan, 2019). Tang et al. (2020). used model information to estimate the states of a shaking table and the constitution of the full state feedback control, which made full use of the known model information, but its performance was too dependent on the established model . The characteristics of the shaking table change dynamically with the frequency of the input signal, and the identification models of different frequency bands are also different (Plummer, 2007). Additionally, the load variation of a shaking table will affect the system output, those and these factors make it difficult for the above algorithm to accurately estimate the system state. Based on the literature [24], Shao et al. (2021) estimated the state of a shaking table system based on an online identification model and proposed the adaptive control method . This method is a simple, practical, and intuitive way to obtain the system state by using the identification model. However, the above algorithm is affected by noise, model change, and external disturbance in practice, and it has stability problems, which makes it difficult to popularize in practice.

To further improve the performance of ADRC, Chu et al. defined the 2-DOF structure of ADRC as a prefilter and a controller, and they tuned the parameters of the ADRC using a frequency domain method (Chu, Wu, & Sepehri, 2019). However, this method is difficult to use in field parameter setting. For the tuning methods of higher-order systems, He et al. (2020) provided the asymptotic conditions for the parameter tuning rules based on sensitivity constraints. Wang, Zu, et al. (2019) proposed a special ADRC parameter setting method for time-delay systems. However, those methods were improved based on the traditional bandwidth method and inherited its defects. These methods have certain limitations when applied to the control of a shaking table, and the setting of the parameters is difficult. Therefore, it is necessary to study a parameter setting method with strong adjustability and easy generalization.

The low accuracy of the state estimated with ESO and the difficulty with tuning the ADRC parameters greatly limit its application in practice when it is applied to a shaking table. To solve the above problems, this paper proposes an improved LADRC by designing a model-based state estimator and replacing the state observed by ESO with the state estimated by the state estimator to form the control law. This allows ESO to focus on observing the total disturbance of the system.

The main motivation behind this research is to design a control method that can meet the requirements of seismic structure tests under different working conditions in RTHS to reproduce a waveform with high accuracy on shaking tables and reduce the difficulty of setting the parameters. These goals are achieved by: (1) effectively utilizing the model information and introducing an adaptive dynamic compensation algorithm to construct a state estimator named the adaptive-model-based state estimator (AMSE), which can accurately estimate the shaking table state in noise pollution and different loading conditions, and (2) using the accurate state estimated by AMSE and the total disturbance observed by ESO to design the ADRC law for different working conditions, and ensure the control accuracy. The main contributions of this research are as follows:

(1) Considering the nonlinear and external disturbances existing in the modeling process, the shaking table model of earthquake simulation with disturbances is established.

(2) The model information and the adaptive phase–amplitude compensation algorithm are used to construct the AMSE, which can accurately estimate the state of the shaking table in different disturbances by compensating for the amplitude and phase deviation of the estimated state, and this has a good filtering effect.

(3) AMSE-ADRC is formed by combining the designed AMSE with ADRC to improve the control accuracy and and adapt the suppressed influence of the shaking table output under different working conditions and specimens.

(4) The pole assignment bandwidth method is proposed based on the original bandwidth method and the separation principle to adjust the control parameters. Additionally, the parameters set by the simulation can be directly used in the actual test without stability problems.

2. Shaking table modeling

A shaking table is an important part of RTHS that is used to simulate seismic waves and achieve high-precision seismic wave recurrence. This section describes how the state-space model of the shaking table with nonlinear action is established. The shaking table system is composed of an actuator unit, hydraulic oil source unit, control unit, data acquisition unit, and support unit. The hydraulic schematic diagram of the shaking table system is shown in Fig. 1. In Fig. 1, q_i and q_o represent the flows into and out of the hydraulic cylinder, respectively, p_i and p_o represent the pressures of the oil inlet and outlet cavities of the hydraulic cylinder, respectively, A_p is the effective area of the piston of the hydraulic cylinder, x_v represents the spool displacement, and y is the displacement of the piston rod, and V_t is the total volume of the oil inlet and outlet cavities of the hydraulic cylinder.

Considering the influence of the pressure pulsation of the hydraulic oil source, the compressibility of the oil in the servo valve, the leakage of the oil cylinder, and the friction of the hydraulic components, the

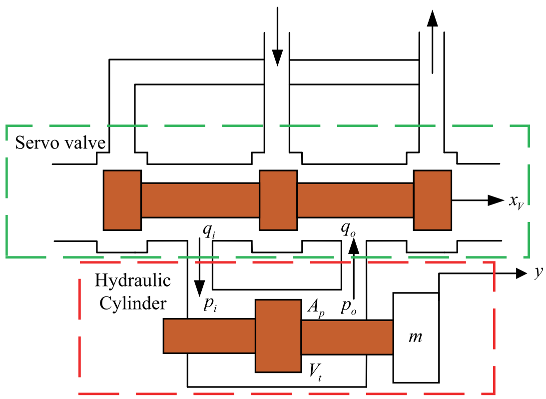

Fig. 1. Hydraulic schematic diagram of shaking table.

flow equation of the electric servo valve and the flow equation of the hydraulic cylinder can be established based on Fig. 1, as follows,

$$q_L = K_q x_v - K_c \left(p_L + \Delta p_L(t)\right) = K_q x_v - K_c p_L - K_c \Delta p_L(t) \tag{1}$$

$$q_L = A_p \frac{dy}{dt} + \frac{V_t}{4\beta_e}\frac{dp_L}{dt} + C_{ic} p_L + \frac{V_t}{4\beta_e}\frac{d\Delta p_L(t)}{dt} \tag{2}$$

where, $q_L = q_i - q_o$ represents the load flow, K_q and K_c are the servo valve flow gain and the flow-pressure coefficient, $p_L = p_i - p_o$ is the load pressure drop, and $\Delta p_L(t)$ is the influence of the pressure pulsation and oil compression on the oil pressure, C_{ic} represents the leakage coefficient of the hydraulic cylinder, and β_c is the effective volumetric elastic modulus.

Considering the Coulomb friction of the system and the viscous properties of the load, the force equilibrium equation can be established according to Newton's second law,

$$F = A_p p_L = m\frac{d^2y}{dt^2} + B_P\frac{dy}{dt} + F_L + F_\mu \tag{3}$$

where F is the driving force of the hydraulic cylinder. m is the total mass converted by the piston and the external load. B_p is the viscous damping coefficient of the piston and the load. F_L is the load force generated by the vibration of the external specimen on the hydraulic cylinder, and the force that is generated varies with the order and parameters of the specimen. F_μ is the internal friction force of the system. A Laplace transform is applied to Eqs. (1), (2) and (3) to obtain the following equations,

$$Q_L = K_q X_v - K_c P_L - \overbrace{\Delta K_c(s) P_L}^{\Delta Q_{L1}} \tag{4}$$

$$Q_L = A_p sY + \frac{V_t}{4\beta_e} sP_L + \overbrace{C_{ic} P_L}^{\Delta Q_{L2}} \tag{5}$$

$$A_p P_L = ms^2 Y + B_P sY + F_L + F_\mu \tag{6}$$

The structure block diagram of the actuator system can be derived from the Eqs. (4), (5) and (6)

According to Fig. 2, the relationship between the piston rod displacement Y, servo valve displacement X_v, load force F_L and disturbances can be deduced

$$Y = \frac{K_1 X_v}{\frac{s^3}{\omega_n^2} + \frac{2\xi_n}{\omega_n}s^2 + s} - \frac{K_2\left(\Delta Q_{L1} + \Delta Q_{L2}\right)}{\frac{s^3}{\omega_n^2} + \frac{2\xi_n}{\omega_n}s^2 + s} - \frac{K_3\left(\frac{s}{\omega_f} + 1\right)\left(F_L + F_\mu\right)}{\frac{s^3}{\omega_n^2} + \frac{2\xi_n}{\omega_n}s^2 + s} \tag{7}$$

where, $K_1 = \frac{A_P K_q}{A_p^2 + K_c B_p}$ represents the displacement gain of the spool,

$K_2 = \frac{A_P}{A_p^2 + K_c B_p}$, represents the modeling error gain,

$K_3 = \frac{K_c}{A_p^2 + K_c B_p}$, represents the gain of the frictional resistance and external load,

$\omega_f = \frac{4\beta_c K_c}{V_t}$, represents the corner frequencies,

$\omega_n = \sqrt{\frac{4\beta_e\left(A_p^2 + K_c B_p\right)}{V_t m}}$, represents the natural hydraulic frequency of the system, and

$\xi_n = \left(K_c + \frac{V_t B_P}{4\beta_e}\right)\sqrt{\frac{\beta_e}{V_t m\left(A_p^2 + K_c B_p\right)}}$, represents the hydraulic damping ratio of the system.

In Eq. (7), three subitems are contained in the right term. The first term represents the displacement of the piston rod of the hydraulic cylinder when it is unloaded, the second term represents the influence of the unmodeled dynamics on the displacement, and the third term represents the influences of the external load and friction on the system. The mathematical model of the servo valve can be expressed as the second-order oscillation loop,

$$G_V(s) = \frac{Q}{I} = \frac{K_V}{\frac{s^2}{\omega_V^2} + \frac{2\xi_V}{\omega_V}s + 1} \tag{8}$$

where Q, I, K_V, ξ_V, and ω_V represent the no-load flow, input current of the electro-hydraulic servo valve, flow gain, damping ratio, and natural frequency of the electro-hydraulic servo valve, respectively. According to Eqs. (7) and (8), the structural block diagram of the shaking table can be obtained as shown in Fig. 3. K_a and K_f are the servo amplification gain and the displacement sensor gain, respectively.

In general, Fig. 3 shows a unit feedback, that is, $K_f = 1$. The output of the position servo system of the system is expressed as

$$Y = \frac{K_4\left(R - D - S_V\right)}{\left(\frac{s^3}{\omega_n^2} + \frac{2\xi_n}{\omega_n}s^2 + s\right) + K_4} \tag{9}$$

where $K_4 = \frac{K_a K_V K_1}{A_p}$

$$D = \frac{1}{K_5}\left(\frac{s^2}{\omega_h^2} + \frac{2\xi_h}{\omega_h}s + 1\right)\left(K_3\left(\frac{s}{\omega_f} + 1\right)\left(F_L + F_\mu\right) + K_2\left(\Delta Q_{L1} + \Delta Q_{L2}\right)\right) \tag{10}$$

and

$$S_V = \frac{2\xi_v}{K_5\omega_v}s^2\left(\frac{s}{2\xi_v\omega_v} + 1\right)\left(\frac{s^2}{\omega_n^2} + \frac{2\xi_n}{\omega_n}s + 1\right)Y \tag{11}$$

The state space equation of the system can be obtained from Eq. (9),

$$\begin{cases} \dot{\mathbf{x}} = A\mathbf{x} + Bu + Bd \\ y = C\mathbf{x} \end{cases} \tag{12}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -K_4\omega_n^2 & -\omega_n^2 & -2\xi_n\omega_n \end{bmatrix} B = \begin{bmatrix} 0 \\ 0 \\ K_4\omega_n^2 \end{bmatrix} C = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \tag{13}$$

$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$, where x_1, x_2 and x_3 are the position, velocity, and acceleration of the actuator, respectively. d is the modeling error, which includes the D and S_V in Eq. (9), and other disturbances exist in the system. ω_n and ξ_n are the natural frequency and the equivalent damping ratio of the shaking table system, respectively. The parameters shown in Eq. (12) are obtained with the subspace model identification method.

Remark 1. Under ideal conditions, the friction, hydraulic oil leakage, external load and other nonlinear factors are not considered in the modeling. Then $d = 0$ and the established system is an idealized model. In Eq. (12), d represents the influence of the external load, nonlinearity and modeling error on the idealized system. In this research, d is

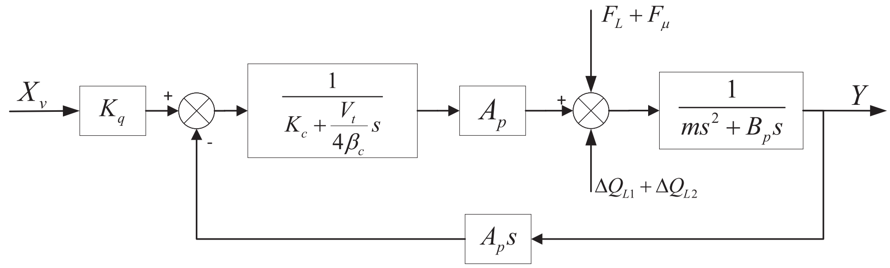

Fig. 2. The structure block diagram of the actuator system.

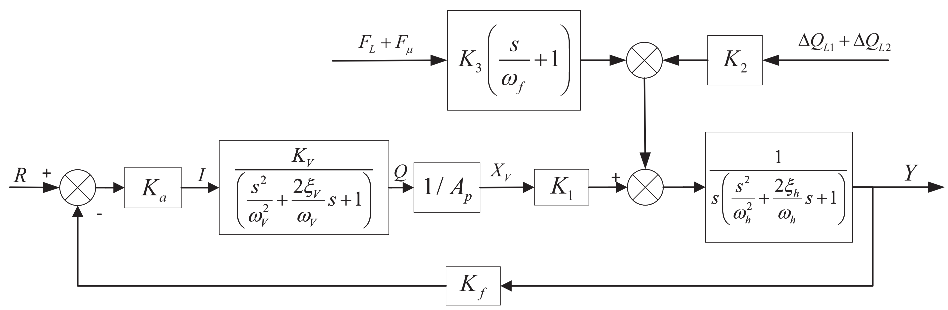

Fig. 3. Shaking table structure block diagram.

regarded as a disturbance, so Eq. (12) can represent the equation of state of the shaking table system under the influence of different external disturbances and different frequency band characteristics.

3. Controller design

In this section, a model-based adaptive state estimator (AMSE) is designed to use the complete modeling information and obtain more accurate state estimates. The ADRC is improved by using the estimated states, and the AMSE-ADRC method is proposed.

3.1. AMSE design

A general third-order system with unmodeled dynamic, nonlinear, and external disturbances is considered, as follows

$$G(s) = \frac{b(1+\Delta(s))}{s^3 + a_3 s^2 + a_2 s + a_1} \tag{14}$$

where $\Delta(s)$ represents the unmodeled dynamic, nonlinear, and external disturbances contained in the system. b, a_1, a_2 and a_3 are model parameters corresponding to Eq. (9). Throughout this research, the system shown in Eq. (14) is assumed to satisfy the following assumptions.

Assumption 1. The controlled object shown in Eq. (14) is stable, that is $a_n > 0$, $n = 1, 2, 3$, and $a_2 a_3 - a_1 > 0$.

Assumption 2. The unmodeled dynamic, nonlinear and external disturbances contained in the system are bounded, that is, there is a positive number $Q \in R^+$, so

$$\left| \frac{\Delta(s)}{s^3 + a_3 s^2 + a_2 s + a_1} \right| < Q \tag{15}$$

The system identification model shown in Eq. (16) is obtained with the subspace identification method

$$G_{id}(s) = \frac{b}{s^3 + a_3 s^2 + a_2 s + a_1} \tag{16}$$

Generally speaking, under the influence of $\Delta(s)$, the stability of the actual system shown in Eq. (14) is worse than that of the identification model shown in Eq. (16). That is, compared with the identification model, the phase lag and amplitude increase with the change of the frequency band that exists in the actual system when drawing the bode diagram. In order to estimate the state of the system more accurately, make full use of the model information, and avoid the stability problems existing in the actual system, the adaptive compensation network is designed to fit the phase and amplitude deviations between Eqs. (14) and (16), as follows,

$$G_\tau(s) = k_\tau \frac{1+\kappa s}{1+\gamma s} \tag{17}$$

where k_τ and γ are the amplitude and phase adaptive parameters, respectively. κ represents the undetermined constants, $\gamma, \kappa, k_\tau > 0$, and $\gamma > \kappa$. κ is used to balance the stability of the system, so that the range of differential action covers the working bandwidth of the system in the frequency domain, and generally, $\kappa = 0.0005$. The AMSE is formed by the adaptive compensation loop shown in Eq. (17) and the identification model shown in Eq. (16), as shown in Eq. (18),

$$G_m(s) = G_{id}(s)\, G_\tau(s) \tag{18}$$

By adjusting the parameters k_τ and γ and fixing the parameter κ, the amplitude and phase of the system change accordingly, as shown in Fig. 4. With the increase of the parameter k_τ, the logarithmic amplitude–frequency characteristics of the system shift upward, while the phase-frequency characteristics are not affected, which means that the parameter k_τ is able to compensate for the deviation of the amplitude–frequency characteristics deviation between the identification model and the actual system without affecting the phase. With the increase of the parameter γ, the high-frequency band of the log-phase-frequency characteristic curve and the log-amplitude–frequency characteristic curve of the system decreases overall, while the low-frequency band of the log-amplitude–frequency characteristic curve changes only slightly, which indicates that the parameter γ can effectively correct the phase of the model, and the larger γ is, the stronger the adjustment effect is.

The state-space equations of the actual system and AMSE can be obtained with Eqs. (14) and (18), as shown in Eqs. (19) and (20), respectively,

$$\begin{cases} \dot{\boldsymbol{v}} = A_v \boldsymbol{v} + B_v u + E\Delta\left(x_1, x_2, x_3, h(t)\right) \\ y_v = C_v \boldsymbol{v} \end{cases} \tag{19}$$

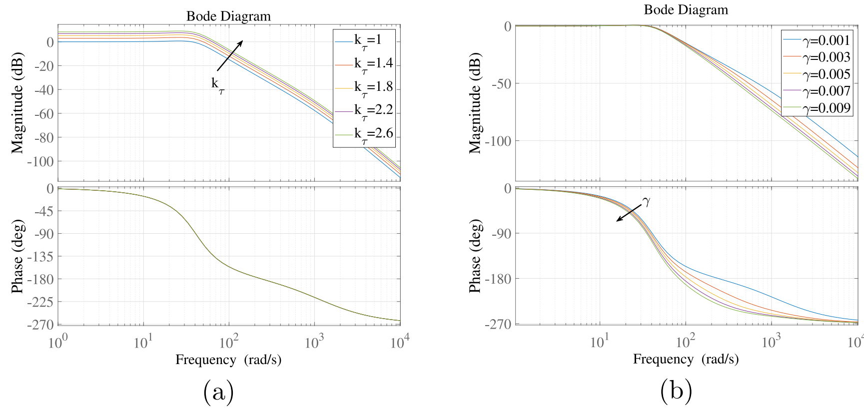

Fig. 4. Open loop bode diagram of the system with adaptive parameter change.

$$\begin{cases} \dot{\boldsymbol{v}}_{id} = A_v \boldsymbol{v}_{id} + B_v u + E\delta \\ y_{id} = C_v \boldsymbol{v}_{id} \end{cases} \tag{20}$$

where $\boldsymbol{v} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$, $\boldsymbol{v}_{id} = \begin{bmatrix} x_{id1} & x_{id2} & x_{id3} \end{bmatrix}$. $\boldsymbol{v}_{id}$ is the identification state vector of the actual system. $\Delta\left(x_1, x_2, x_3, h(t)\right) = c_1x_1 + c_2x_2 + c_3x_3 + h(t)$, represents $\Delta(s)$ in Eq. (14). δ, as shown in Eq. (22), is the reconstruction of $\Delta\left(x_1, x_2, x_3, h(t)\right)$. For the design of $G_\tau(s)$, δ accurately reconstructs $\Delta(t)$ at every moment. Then, the vector $\boldsymbol{v}_{id}$ converges to $\boldsymbol{v}$.

$$A_v = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} B_v = \begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix} E = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} C_v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \tag{21}$$

$$\begin{cases} \delta = n_1 x_{id1} + n_2 x_{id2} + n_3 x_{id3} + n_4 \dot{x}_{id3} + n_5 u + n_6 \dot{u} \\ J = \frac{1}{\gamma a_3 + 1} \\ n_1 = J\gamma a_3 a_1 \\ n_2 = J\gamma\left(a_3 a_2 - a_1\right) \\ n_3 = J\gamma\left(a_3^2 - a_2\right) \\ n_4 = -J\gamma \\ n_5 = b\left(Jk_\tau - 1\right) \\ n_6 = Jk_\tau b\lambda \end{cases} \tag{22}$$

The parameters γ and k_τ are adaptively tuned using Eq. (23)

$$\begin{cases} \gamma = \gamma_0 + \sigma_\tau \tanh(\dot{r})(r - y) \\ k_\tau = k_{\tau 0} + \sigma_k \tanh(\ddot{r})(r - y) \end{cases} \tag{23}$$

where σ_τ and σ_k are the gains of the delay and amplitude parameters, respectively, and the larger the values of the gains are, the faster the convergence rate is, but the more likely the estimated value is to oscillate. γ_0 and $k_{\tau 0}$ are the values of γ and k_τ to be updated. The initial values of γ_0 and $k_{\tau 0}$ are κ and 1.

Remark 2. For the design of $G_\tau(s)$, the adaptive tuning parameters are γ and k_a, the phase and amplitude characteristics of the identification system are adjusted, and δ is used to accurately reconstruct $\Delta(t)$ at every moment so that the AMSE can accurately estimate the actual system state, with $\boldsymbol{v}_{id} = \boldsymbol{v}$. Since the parameters σ_τ and σ_k generally take small values, the noise in the feedback signal y is weakened, which causes the AMSE to have a better filtering effect.

3.2. AMSE-ADRC design

Let $x_4 = -a_1x_1 - a_2x_2 - a_3x_3 + E\Delta(t)$ and $\boldsymbol{\eta} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}$, where x_4 represents the expanded state and the total disturbance of the system, including the modeling error, nonlinear characteristics, and external disturbances. Then Eq. (19) can be rewritten as

$$\begin{cases} \dot{\boldsymbol{\eta}} = A_\eta \boldsymbol{\eta} + B_\eta u \\ y_\eta = C_\eta \boldsymbol{\eta} \end{cases} \tag{24}$$

where

$$A_\eta = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} B_\eta = \begin{bmatrix} 0 \\ 0 \\ b \\ 0 \end{bmatrix} C_\eta = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}^T \tag{25}$$

In order to incorporate all of the model information, an AMSE-based ESO is designed for Eq. (24)

$$\begin{cases} \dot{\hat{\mathbf{x}}} = A_\eta \hat{\mathbf{x}} + B_\eta u + L(y_{id} - \hat{y}) \\ \hat{y} = C_\eta \hat{\mathbf{x}} \end{cases} \tag{26}$$

where $\hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 & \hat{x}_4 \end{bmatrix}$, and $L = \begin{bmatrix} l_1 & l_2 & l_3 & l_4 \end{bmatrix}$. $\hat{x}_1$, $\hat{x}_2$, $\hat{x}_3$, and $\hat{x}_4$ are the estimated displacement, velocity, acceleration and total disturbance respectively. l_1, l_2, l_3, and l_4 are the undetermined parameters of the ESO. L is tuned with the traditional bandwidth method (Gao, 2013), that is $l_1 = 4\omega_o$, $l_2 = 6\omega_o^2$, $l_3 = 4\omega_o^3$, and $l_4 = \omega_o^4$, where ω_o is the observer bandwidth. Based on the observed total disturbance, the active disturbance rejection control law is designed as shown in Eq. (27),

$$u = \frac{u_0 - \hat{x}_4}{b} \tag{27}$$

By substituting Eq. (27) into Eq. (19) and eliminating the total disturbance of the system through the ESO, a third-order integral series type without a disturbance can be obtained

$$\begin{cases} \dot{\boldsymbol{v}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \boldsymbol{v} + \begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix} u_0 \\ y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \boldsymbol{v} \end{cases} \tag{28}$$

The state feedback control is generally used for the integral series control shown in Eq. (28), which can achieve an ideal control effect. State feedback control is built from the state estimated by the AMSE rather than that of ESO, which can make full use of the known model information and is the big difference between the improved ADRC and the original ADRC. The PDD control law composed of the state estimated by the AMSE is shown in Eq. (29). The matrix expression of the control law in Eq. (30) can be obtained by combining Eq. (27) and Eq. (29). The extended state model in Eq. (24), ESO in Eq. (26), AMSE in Eq. (20) and the improved PD control law (30) constitute AMSE-ADRC, whose structure diagram is shown in Fig. 5

$$u_0 = k_p\left(r - x_{id1}\right) - k_d x_{id2} - k_{dd} x_{id3} \tag{29}$$

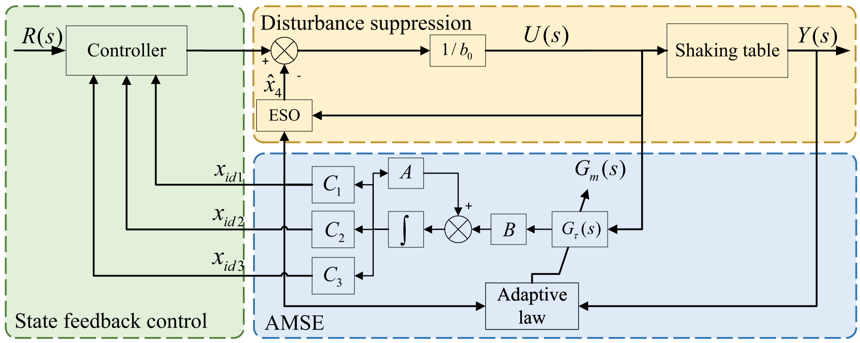

Fig. 5. Structure diagram of AMSE-ADRC.

$$u=\frac{K_cCr-K_c\boldsymbol{\upsilon}_{id}-M\hat{\mathbf{x}}}{b} \tag{30}$$

where

$$\begin{aligned} K_c &= \begin{bmatrix} k_p & k_d & k_{dd} \end{bmatrix} \\ M &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{31}$$

Remark 3. Since the AMSE takes into account all of the dynamic characteristics of the shaking table, the AMSE-ADRC parameters obtained based on the simulation can be directly used in experiments.

4. Stability and separation principle

This section describes how the stability condition of the closed-loop system and the separation principle of the ESO, AMSE, and ADRC controllers are proven by the Hurwitz criterion. Next, the pole assignment parameter setting method of the ADRC controller is proposed.

4.1. Stability analysis

Eq. (30) is substituted into Eq. (19) to obtain

$$\begin{cases} \dot{\boldsymbol{\upsilon}}=A_\upsilon\boldsymbol{\upsilon}+\frac{B_\upsilon}{b}\left(K_cCr-K_c\boldsymbol{\upsilon}_{id}-M\hat{\mathbf{x}}\right)+E\Delta\left(x_1,x_2,x_3,h(t)\right) \\ y_\eta=C_\upsilon\boldsymbol{\upsilon} \end{cases} \tag{32}$$

It is assumed that

$$\mathbf{e}_{id}=\boldsymbol{\upsilon}-\boldsymbol{\upsilon}_{id} \tag{33}$$

$$\mathbf{e}_{eso}=\boldsymbol{\eta}-\hat{\mathbf{x}} \tag{34}$$

$$\chi=\begin{bmatrix} \boldsymbol{\upsilon} & \mathbf{e}_{eso} & \mathbf{e}_{id} \end{bmatrix} \tag{35}$$

Based on Eqs. (33) and (34), Eq. (32) can be rewritten as Eq. (36). Combining Eqs. (19) and (20), the identification error shown in Eq. (37) is established. With Eqs. (24) and (26), the ESO estimation error shown in Eq. (38) is obtained.

$$\dot{\boldsymbol{\upsilon}}=A_k\boldsymbol{\upsilon}+E\left(K_c\mathbf{e}_{id}+M\mathbf{e}_{eso}\right)+EK_cCr \tag{36}$$

$$\dot{\mathbf{e}}_{id}=\left(A_\upsilon+E\mathbf{N}\right)\mathbf{e}_{id}+E\left(\Delta_1\left(x_1,x_2,x_3,h(t)\right)-\delta_1\right) \tag{37}$$

$$\dot{\mathbf{e}}_{eso}=\left(A_\eta-LC_\eta\right)\mathbf{e}_{eso}+LC\mathbf{e}_{id} \tag{38}$$

where

$$\Delta_1\left(x_1,x_2,x_3,h(t)\right)=\left(c_1-n_1\right)x_1+\left(c_2-n_2\right)x_2+\left(c_3-n_3\right)x_3+h(t) \tag{39}$$

$$\delta_1=n_4\dot{x}_{id3}+n_5u+n_6\dot{u} \tag{40}$$

$$A_k=\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_p & -k_d & -k_{dd} \end{bmatrix}\mathbf{N}=\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}^T \tag{41}$$

Combined with Eqs. (36), (37), and (38), there is

$$\dot{\chi}=A_\chi\chi+\overbrace{EK_cCr+E_\chi\left(\Delta_1\left(x_1,x_2,x_3,h(t)\right)-\delta_1\right)}^{\Psi} \tag{42}$$

where

$$A_\chi=\begin{bmatrix} A_k & EM & EK_c \\ \mathbf{0} & A_\eta-LC_\eta & LC \\ \mathbf{0} & \mathbf{0} & A_\upsilon+E\mathbf{N} \end{bmatrix}_{10\times10} \quad E_\chi=\begin{bmatrix} \mathbf{0}_{3\times1} \\ \mathbf{0}_{4\times1} \\ E \end{bmatrix}_{10\times1} \tag{43}$$

Lemma 1. *For the system shown in Eq. (31), when* $\mathbf{T}$ *is the Hurwitz matrix and* $g(\theta)$ *is bounded, the system is bounded input bounded output (BIBO) stable. Gao (2006)*

$$\dot{\theta}=\mathbf{T}\theta+g(\theta) \tag{44}$$

Theorem 1. *According to Assumptions 1 and 2,* Ψ *is further assumed to be bounded, that is, there is a positive number* m *that makes* $|\Psi|<m$*. Then the design of Eqs. (20), (26), and (30) is BIBO-stable closed loop if* A_χ *is Hurwitz, that is*

(1) A_k *is Hurwitz, and*
(2) $A_\eta-LC_\eta$ *is Hurwitz, and*
(3) $A_\upsilon+E\mathbf{N}$ *is Hurwitz.*

The eigenvalue of A_χ is located in the left half plane of the s field and obtained with Eqs. (45), (46), and (47)

$$\det\left(\lambda I-A_k\right)=0 \tag{45}$$

$$\det\left(\lambda I-A_\eta+LC_\eta\right)=0 \tag{46}$$

$$\det\left(\lambda I-A_\upsilon-E\mathbf{N}\right)=0 \tag{47}$$

The appropriate controller coefficient K_c, observer coefficient L, and adaptive parameter γ are selected so that A_k, $A_\eta-LC_\eta$ and $A_\upsilon+E\mathbf{N}$ are Hurwitz matrixes. Then the eigenvalues obtained with Eqs. (45), (46), and (47) are all located in the left half plane of the s domain. The characteristic equation of the AMSE can be obtained with the expansion of Eq. (47),

$$\lambda^3+\frac{a_3+\gamma a_2}{\gamma a_3+1}\lambda^2+\frac{a_2+\gamma a_1}{\gamma a_3+1}\lambda+\frac{a_1}{\gamma a_3+1}=0 \tag{48}$$

According to the Routh criterion, when the eigenvalue of Eq. (47) is in the left half plane of the domain, the following must be satisfied

$$\begin{cases} \gamma>0 \\ a_2a_3+a_1a_2\gamma^2+a_2^2\gamma>a_1 \end{cases} \tag{49}$$

4.2. Pole assignment bandwidth method

The characteristic equation of the controller can be obtained with the expansion of Eq. (45),

$$s^3 + k_{dd}s^2 + k_d s + k = 0 \tag{50}$$

The traditional bandwidth method assigns poles at the same point $-\omega_c$ on the real axis of the left half plane of the s domain, which limits the performance of the controller. In this research, pole assignment is adopted to set the controller parameters, and the system poles are assigned at the points $-p_1\omega_c$, $(-1+q_2 i)\omega_c$ and $(-1-q_2 i)\omega_c$ respectively, which yields the following,

$$\begin{aligned} &s^3 + k_{dd}s^2 + k_d s + k = \\ &\left(s + p_1\omega_c\right)\left[s + \left(1 + q_2 i\right)\omega_c\right]\left[s + \left(1 - q_2 i\right)\omega_c\right] \end{aligned} \tag{51}$$

and by comparing the coefficient, the controller parameters can be obtained as follows

$$\begin{cases} k_p = p_1\left(1 + q_2^2\right)\omega_c^3 \\ k_d = \left(1 + q_2^2 + 2p_1\right)\omega_c^2 \\ k_{dd} = \left(2 + p_1\right)\omega_c \end{cases} \tag{52}$$

where $p_1, q_2, \omega_c \in R^+$. The parameters p_1 and q_2 change the real and imaginary axis poles of the controller, respectively. The traditional bandwidth method can be regarded as a special case of the pole assignment bandwidth method. In the traditional ADRC control law, the weights of the displacement, velocity, and acceleration in the control law are fixed, but the pole assignment bandwidth method makes the weight adjustable. By changing the virtual axis pole position, the influence of the displacement on the control law can be increased, the influence of the velocity and acceleration on the control law can be reduced, and the influence of the noise on the control effect can be reduced.

The parameter setting takes the traditional bandwidth method as the starting point by default. When setting ω_c at $p_1 = 1$ and $q_2 = 0$ fails to achieve the best effect, q_2 is increased and then ω_c is adjusted repeatedly. If this is still not ideal, p_1 is increased and then ω_c is adjusted repeatedly.

Remark 4. Eqs. (45)–(47) represent the eigenvalue of the controller, ESO, and AMSE, respectively. The above equation shows that the designs of the ESO, AMSE, and ADRC control laws satisfy the separation principle, and the stability of the whole system is ensured by the separate design of each part. The pole assignment bandwidth method is proposed based on the above separation principle. The controller poles can be assigned by tuning the parameters p_1 and q_2. The pole assignment bandwidth method is equivalent to the traditional bandwidth method when $p_1 = 1$ and $q_2 = 0$. As can be seen from Eq. (53), increasing p_1 or q_2 can increase the weight of the displacement signal in the control law.

5. Simulation and experiment

In this section, verification of the proposed AMSE-ADRC method is discussed based on simulations and experiments on a shaking table. Firstly, the effectiveness of the AMSE was tested on a shaking table model by comparing the results of the linear ESO and AMSO. Then, the simulation results of the AMSE-ADRC method verified the adaptability to different working conditions and the robustness to disturbances. Finally, based on the simulation parameters, AMSE-ADRC was applied on a shaking table with different specimens and working conditions. The advantages of AMSE-ADRC in the setting of the parameters, and the control accuracy and robustness of the algorithm were verified by comparison with the existing shaking table control algorithms.

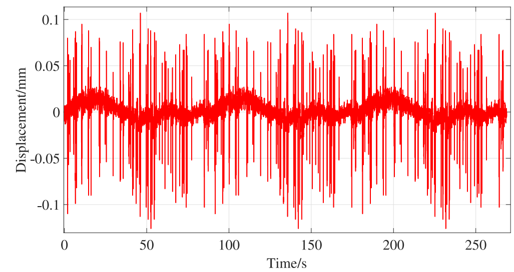

Fig. 6. Displacement sensor noise for shaking table.

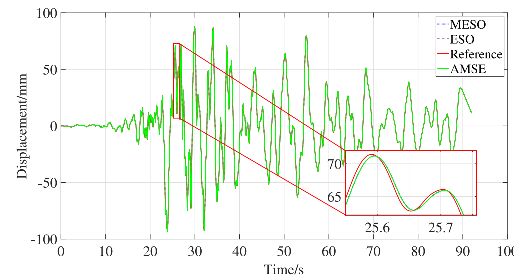

Fig. 7. Displacement comparison between MESO, AMSE, and ESO for electro-hydraulic shaking tables.

5.1. Simulation test

Based on the established model shown in Eq. (9), the subspace model identification method is used to identify the shaking table model. The natural frequency of the shaking table is $\omega_n = 237.88$, and the identification parameters are $K = 36.81$ and $\xi_n = 2.66$. The system matrix of the shaking table is obtained, as shown in Eq. (53),

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2083000 & -56590 & -1267 \end{bmatrix} \\ B &= \begin{bmatrix} 0 & 0 & 2083000 \end{bmatrix}^T \end{aligned} \tag{53}$$

The AMSE-ADRC is designed for the model, and the controller parameters are set as: $\omega_c = 80$, $\omega_o = 400$, $b = 400000$, $p_1 = 3$, $q_2 = 5$, $\kappa = 0.0005$, $\sigma_\tau = 0.000001$, and $\sigma_k = 0.00001$. At the same time, the conventional ADRC controller and the GADRC proposed in Fu and Tan (2016) are designed as the control simulation test, and their parameters are set as: $\omega_c = 90$, $\omega_o = 270$, and $b = 400000$. The sampling period of the control system is 0.001 s. Considering the noise interference in the actual test, the displacement sensor noise collected as shown in Fig. 6 is added to the output of the simulation model in this research.

Figs. 7–9 show the comparison of the model-based ESO (MESO), AMSE, and ESO observations of the displacement, velocity, and acceleration signals when the Hokkaido seismic waves are reproduced by electro-hydraulic shaking tables, which are controlled by GADRC, AMSE-ADRC, and conventional LADRC. Fig. 10 displays the frequency spectrum comparison of the acceleration obtained with these three algorithms.

As can be seen from Figs. 7–9, it is clear that (1) the AMSE can accurately estimate the state of the shaking table, which means that the position, velocity, and acceleration curves obtained with the AMSE are very close to the actual state and the noise is well filtered. (2) The ESO can accurately observe the displacement and the velocity state. However, there is a large overshoot when the acceleration signal is observed. (3) The MESO can accurately observe the displacement, velocity, and acceleration of the shaking table, but the observed velocity is slightly oscillating due to noise, while the observed acceleration is obviously noise-polluted.

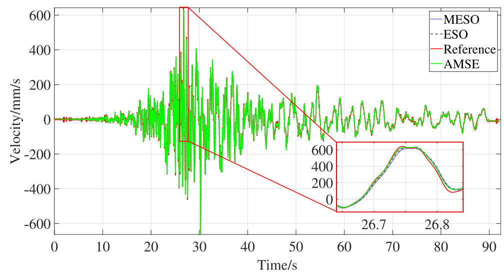

Fig. 8. Velocity comparison between MESO, AMSE, and ESO for electro-hydraulic shaking tables.

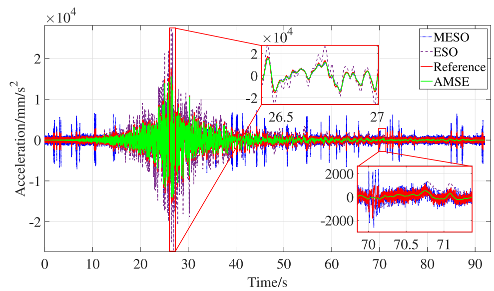

Fig. 9. Acceleration comparison between MESO, AMSE, and ESO for electro-hydraulic shaking tables.

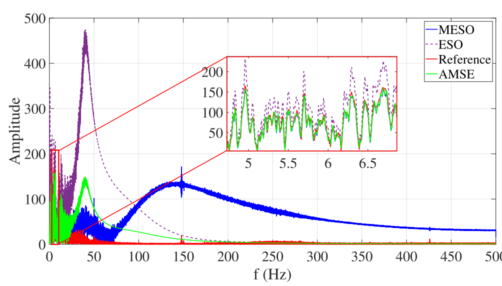

Fig. 10. Spectral comparison of the observed accelerations.

The overshoot and noise pollution problem of acceleration estimation is shown in Fig. 10. As can be seen from the figure, the acceleration observed by the ESO has a large overshoot, and the observed signal deviates greatly from the real value of the system within the range of 25 Hz–100 Hz because of the sensor noise. Therefore, the observed acceleration signal is significantly distorted. The acceleration observed with MESO is close to the real value of the system below 25 Hz, but it is too sensitive to noise, resulting in a large number of burrs in the signal, which greatly deviates from the real value in the high-frequency band. The AMSE can suppress the noise of the observed signal in the high-frequency band, is less affected by the noise, and can accurately reproduce the acceleration signal of the system.

When the distorted state signal is used in the actual shaking table state feedback, the controller parameters are difficult to set, which makes it difficult for the algorithm to achieve the ideal control effect. Replacing the observed state of the ESO with the state estimated by the AMSE can effectively improve the control performance and reduce the difficulty of setting the parameters. In order to better evaluate the performance of the controller, three important evaluation indexes are introduced, which are the time delays between the expected input and the actual output (J_1), root mean square error (J_2), and peak error (J_3), as shown in Table 1. Specifically, the time delay is defined as the phase leading between the output and input.

Table 1
Evaluation indexes of shaking table control.

Criterion	Equation	Unit
Time delay	$J_1 = \arg\max_k \left[\sum_i y_n^{(1)}(i) x_m(i-k)\right]^{-1}$	ms
Root mean square error	$J_2 = \sqrt{\dfrac{\sum_{i=1}^N \left[x_m(i) - y_n^{(1)}(i)\right]^2}{\sum_{i=1}^N \left[y_n^{(1)}(i)\right]^2}} \times 100$	%
Peak error	$J_3 = \dfrac{\max\left\|x_m(i) - y_n^{(1)}(i)\right\|}{\max\left\|y_n^{(1)}(i)\right\|} \times 100$	%

Table 2
Evaluation index comparison between AMSE-ADRC, GADRC and original LADRC.

Criterion	AMSE-ADRC	GADRC	LADRC
J_1	0	3	18
J_2	0.13	2.31	5.46
J_3	0.39	8.79	15.20

Table 3
Evaluation index comparison of shaking table control.

Case	Algorithm	J_1	J_2	J_3
	AMSE-ADRC	0	0.57	2.14
Case 1	MSE-ADRC	5	1.89	4.19
	GADRC	5	2.49	6.85
	AMSE-ADRC	0	0.61	1.84
Case 2	MSE-ADRC	6	2.21	4.89
	GADRC	4	2.66	8.41
	AMSE-ADRC	0	0.74	2.79
Case 3	MSE-ADRC	7	2.52	5.58
	GADRC	5	2.82	8.98
	AMSE-ADRC	0	0.89	1.92
Case 4	MSE-ADRC	5	20.04	20.42
	GADRC	4	9,41	12.34
	AMSE-ADRC	0	1.04	3.30
Case 5	MSE-ADRC	5	20.14	20.60
	GADRC	4	9.16	12.22

Table 2 shows the index comparison of the Hokkaido waveform output of the shaking table between GADRC, AMSE-ADRC, and conventional ADRC. As can be seen from Table 2, based on the accurate observation of the state with the AMSE, the control effect of AMSE-ADRC is obviously better than that of GADRC and conventional LADRC. The time delay is well controlled due to the pole assignment bandwidth method.

The influence of the nonlinear characteristics and external disturbances on the system can be intuitively reflected in the output, which makes the actual output of the system deviate from the expected output and introduces amplitude and phase deviations. Table 3 shows the comparison of the control indexes when delay disturbances of 5 ms (Case 1), 6 ms (Case 2), and 7 ms (Case 3) were added to the system output. The indexes are also shown when the amplitude was reduced by 20% (Case 4) and increased by 20% (Case 5) and the amplitude disturbance was applied at the same time as in Case 1. The comparison group is the control index comparison of the system without an adaptive compensation algorithm ($\gamma = 0$, $\kappa = 0$, $k_\tau = 1$), and it is denoted in the table as MSE-ADRC. Figs. 11–13 show the variations of the adaptive parameters γ and k_τ when Case 1–3 were added at the output without amplitude disturbance, respectively. Figs. 14 and 15 show the variations of the adaptive parameters γ and k_τ when Case4 and 5 were added to the system, respectively.

The subfigures labeled (a) and (b) in Figs. 11–15 show the adaptive variation process of the adaptive law for the parameters γ and k_τ, respectively. As can be seen from Figs. 11–13, parameter γ started to change at about 10 s, and the corresponding seismic wave began to vibrate significantly at 10 s. The parameter γ reached the stable value for the first time around 20 s, and the rise time for reaching different stable values was fixed. The stable value increased with the increase in

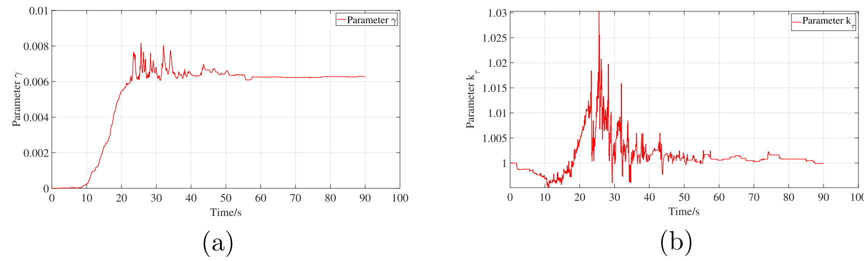

Fig. 11. Change curves of adaptive parameters γ and k_τ with 5 ms delay disturbance.

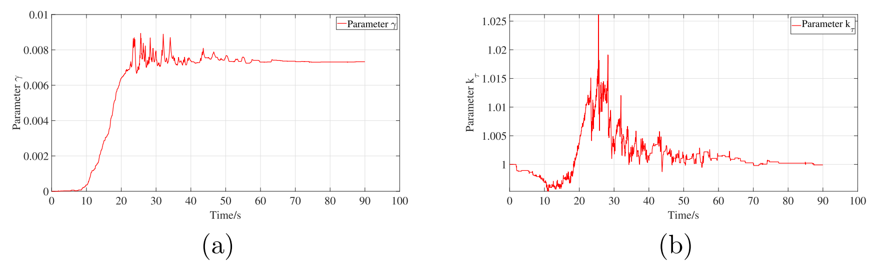

Fig. 12. Change curves of adaptive parameters γ and k_τ with 6 ms delay disturbance.

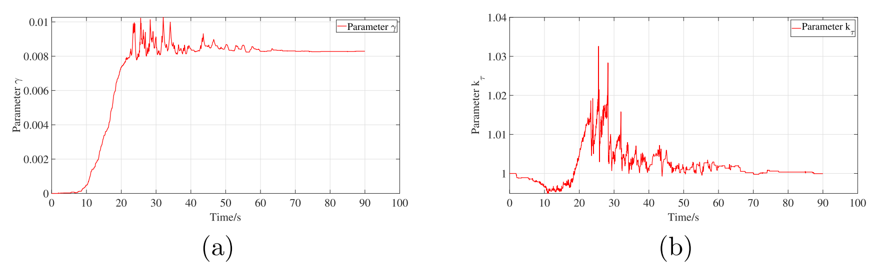

Fig. 13. Change curves of adaptive parameters γ and k_τ with 7 ms delay disturbance.

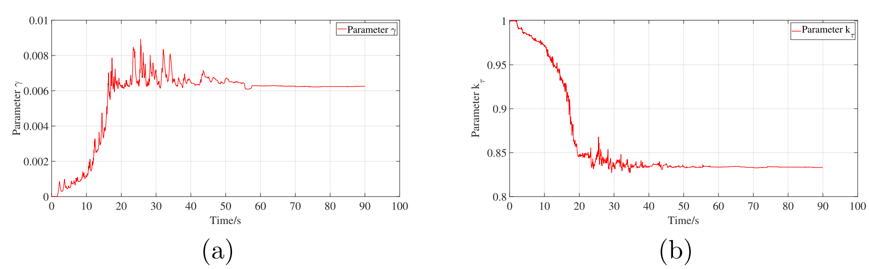

Fig. 14. Change curves of adaptive parameters γ and k_τ when system output amplitude decreases by 20%.

the time-delay disturbance in the system. The tuning parameter γ could effectively fit the delay disturbance of the controlled object.

By comparing the results in Table 3, it can be seen that with the set parameters k_τ and γ, the amplitude disturbances and time delay disturbances of the controlled object could be effectively tracked. There was a slight coupling effect between γ and k_τ, and the addition of k_τ sped up the convergence of γ. The AMSE could adapt well to the output delay and amplitude disturbance of the controlled object, thus achieve a stable control effect. The output delay and amplitude of the identification model were closed to the actual delay and amplitude of the controlled object due to the AMSE.

5.2. Shaking table test

In order to verify the control performance of the AMSE-ADRC method on the shaking table, a shaking table test was designed. The AMSE fully considers the dynamic characteristics of the controlled object such that the simulation parameters of AMSE-ADRC could be directly used in the experiment. The parameters set in this experiment were consistent with the simulation parameters, which were $\omega_c = 70$, $\omega_o = 350$, $b = 400000$, $p_1 = 3$, and $q_2 = 5$, and $\kappa = 0.001$.

To verify the control performance and robustness of AMSE-ADRC in combination with the application requirements of different test specimens, two different seismic waveforms and seven different controlled objects are used to carry out control performance tests, further verifying the feasibility and rationality of improving the ADRC. The seven different specimens are: (1) The electrohydraulic actuator does not have a specimen. (2) The distribution beam specimen is placed near the actuator. (3) The actuator is placed in a 1 DOF box. (4) The actuator is placed in a 1 DOF box, and the box weight is increased by 50 kg. (5) The actuator is placed in a 2-DOF box. (6) The actuator is placed in a 2-DOF box, and the box weight is increased by 50 kg. (7) The actuator is placed in the 2-DOF box, the box weight is increased by 50 kg, and the spring stiffness between the boxes is reduced. The

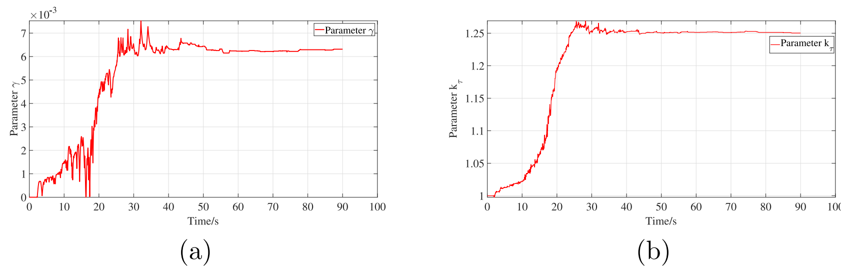

Fig. 15. Change curves of adaptive parameters γ and k_τ when system output amplitude increases by 20%.

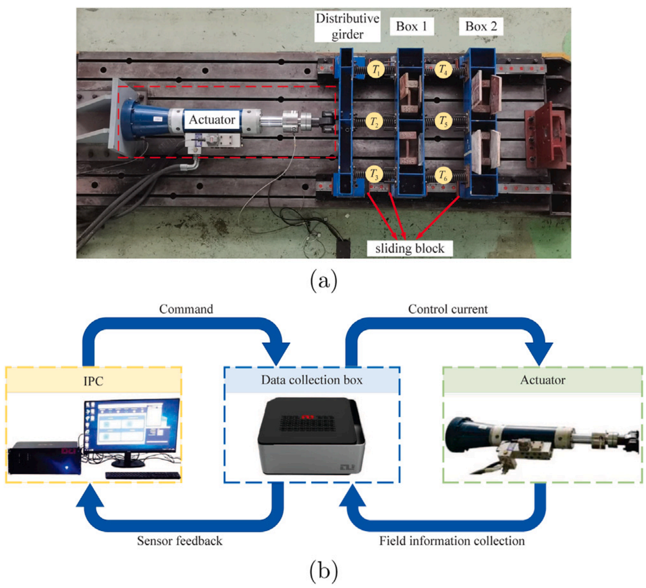

Fig. 16. Shaking table loading site installation diagram.

six types of specimens change the mass, spring stiffness, and order of the specimens, which are used to fully measure the adaptability of the control algorithm to different working conditions. Fig. 16(a) is the loading diagram of the specimen loaded with two degrees of freedom with the box weight increased by 50 kg on the shaking table. Fig. 16(b) shows the field structure of the shaking table control. The designed AMSE-ADRC is embedded in the industrial computer.

In order to verify the influence of the adaptive control compensation in the AMSE-ADRC, the improved ADRC without adaptive compensation, namely MSE-ADRC, is taken as the control test, and its adaptive parameters are set as: $\gamma = 0$ and $k_\tau = 1$. The other parameters of ADRC in the test are consistent with AMSE-ADRC. At the same time, in order to verify the adaptability of AMSE-ADRC to different specimens, only the AMSE-ADRC parameters required by the electrohydraulic actuator without a specimen are set by simulation, and the ADRC parameters will not be set after the replacement of different specimens.

Firstly, an actuator control test is carried out to verify the stability of the control algorithm. The input and output data of the actuator are used to identify the shaking table model. Based on the identification model, the adaptive compensation algorithm is added to form the AMSE, and the control parameter tuning is completed in the simulation. Finally, the simulation algorithm is embedded into the industrial computer to control the actual actuator. To intuitively observe the adaptive compensation effect, a synchronization subspace plot (SSP) is introduced. If there is no time delay, the SSP curve is a straight line with a slope of 1, and if there is a time delay, the SSP curve is elliptical and the width of the ellipse is proportional to the time delay. The comparison of AMSE-ADRC and MSE-ADRC algorithm SSP is shown

Table 4
The coefficients of each spring.

Coefficient	T_1	T_2	T_3	T_4	T_5	T_6
Rigidity (kN/m)	38.5	41.5	39.2	36.9	41.9	37.9
Damp (N s/m)	13	7.2	8.9	7.3	2	5.3

Table 5
Mass of each component.

Component	Sliding block	Distributive girder	Box 1	Box 2
Mass (kg)	9.9	52.3	63.4	63.4

in Fig. 17. Fig. 17(a) and (b) show the SSP diagram of the shaking table reproducing the 004EW and SH000 waveforms controlled by two algorithms. As can be seen from Fig. 17, AMSE-ADRC can control the actuator to accurately output seismic waves with different amplitudes and frequencies by using parameters obtained with the simulation, and its SSP graph is closer to a straight line with a slope of 1, indicating that the adaptive compensation algorithm effectively improves the state estimation ability of the AMSE.

The second step is to load the specimen and test the performance of AMSE-ADRC with six different specimens. The parameters of each part of the box and spring of the specimen are shown in Tables 4 and 5, corresponding to Fig. 16(a). The six kinds of specimens are loaded in turn. The AMSE only uses the model information of the electrohydraulic actuator without specimens, and the system parameters are consistent with the parameters of the first test. The subsequent effects of the loading of six different specimens on the actuator are regarded as disturbances and eliminated by the ESO observation. The SSP comparison diagram of the two algorithms controlling the shaking table loading of different specimens and working conditions is similar to that of Fig. 17, so it is omitted here.

Under the two different working conditions, AMSE-ADRC (denoted as Y in Tables 6–7) and MSE-ADRC (denoted as N in Tables 6–7) control the actuator and six different specimens for the evaluation index data pairs, as shown in Tables 6–7. As can be seen from Tables 6–7, when replacing different specimens, group Y can better control the time delay of the shaking table and the control effect is more stable in comparison to group N. The root mean square error and the peak error are also improved and the control performance is relatively stable. For different working conditions, group Y can also control the time delay stably, while group N has a large delay fluctuation.

Due to the small amplitude of the SHL000 waveform, noise has a relatively large impact on the waveform, and the corresponding mean square error and peak error are also larger than 004EW. The position comparison is shown in Fig. 18. The time delay of the SHL000 waveform for the shaking table is well controlled, and due to the existence of noise, there are some burrs in the waveform, which affects the root mean square and mean value errors.

It can be seen from the above discussion that the time delay, mean square error, and peak error of the shaking table under AMSE-ADRC control are smaller and more stable than those of the improved ADRC

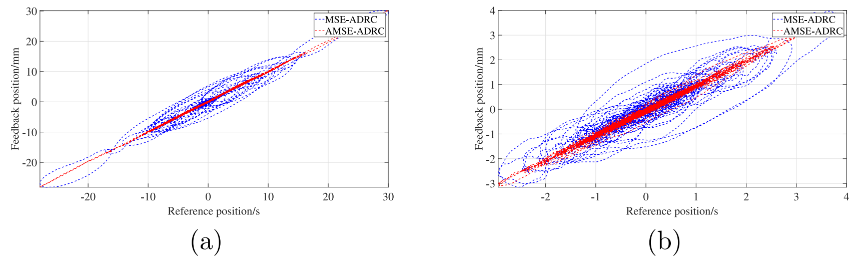

Fig. 17. SSP diagram of the shaking table controlled by two algorithms.

Table 6
Indexes comparison when 004EW is reproduced.

Specimen	Algorithm	J_1	J_2	J_3
Specimen 1	Y	0	2.14	2.08
	N	4	5.43	5.88
Specimen 2	Y	0	2.76	1.92
	N	5	5.47	5.31
Specimen 3	Y	0	2.33	2.89
	N	5	5.66	5.29
Specimen 4	Y	0	2.21	3.41
	N	5	5.60	6.42
Specimen 5	Y	0	2.14	3.53
	N	6	5.56	5.61
Specimen 6	Y	0	2.35	2.83
	N	6	5.50	5.63
Specimen 7	Y	0	2.13	2.85
	N	6	6.20	5.93

Table 7
Indexes comparison when SHL000 is reproduced.

Specimen	Algorithm	J_1	J_2	J_3
Specimen 1	Y	1	10.55	14.43
	N	9	12.31	15.28
Specimen 2	Y	1	9.59	12.01
	N	11	13.18	15.99
Specimen 3	Y	1	12.83	15.46
	N	10	13.41	16.80
Specimen 4	Y	2	10.78	12.77
	N	9	11.90	15.40
Specimen 5	Y	2	9.66	13.88
	N	11	12.99	15.82
Specimen 6	Y	2	9.41	15.17
	N	11	13.39	16.36
Specimen 7	Y	2	9.60	23.36
	N	12	14.42	17.62

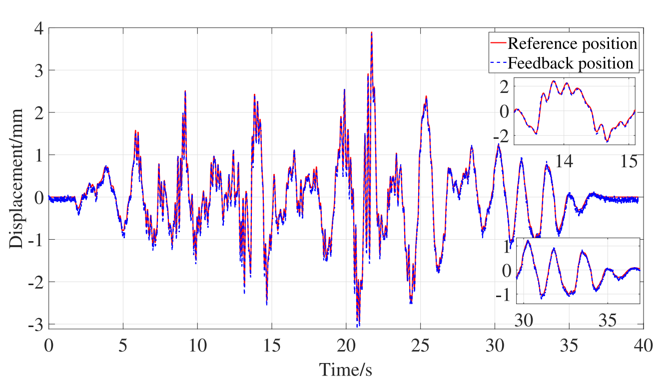

Fig. 18. Displacement comparison of SHL000 reproduced on the shaking table.

control without adaptive compensation. The time delay can be controlled at about 0 ms under different working conditions and specimen loading, and the root mean square error and peak error can also be well controlled.

6. Conclusion

In view of the difficulty with the setting of parameters caused by the inaccurate state estimation of the ESO for a shaking table, this research combines the identification model with the adaptive compensation algorithm to form AMSE, which effectively solves the problems of modeling errors in the identified model and nonlinear system. In this study, to further expand the tuning range of ADRC, the estimated state of the AMSE and the total disturbance estimated by the ESO are used to constitute the control law, and a pole assignment bandwidth method is proposed for tuning the controller parameters and improving the control performance of ADRC. The reasonableness of the proposed AMSE state estimation method, and the stability of AMSE-ADRC are proven using the transfer function derivation and Hurwitz criterion.

Accordingly, a physical test setup for the shaking table is designed, using seven different specimens with two sets of amplitude and frequency configurations for different seismic wave conditions. The test results show that:

(1) The AMSE-ADRC parameters are easy to set, the test parameters are consistent with the simulation parameters, and the control effect is stable.

(2) The AMSE-ADRC design process satisfies the separation principle and enables the system to be stabilized by designing the AMSE, controller, and ESO separately.

(3) The ADRC based on the AMSE and pole assignment significantly improves the control accuracy of the shaking table and has strong robustness to different specimens and working conditions, which means that the replacement of different specimens has little influence on the algorithm's control performance.

The designed algorithm has a good control effect on the external disturbance of the seismic simulation shaking table, but the following aspects still need to be further studied. For example, (1)the main frequency of the test condition in this research is below 10 Hz, which is the frequency range of most seismic waveforms. The test of the high frequency vibration waveform needs to be further studied. (2) The parameters of ESO are still set using the bandwidth method, which limits the estimation effect of ESO on the total disturbance. If the parameter setting method of ESO can be improved, it would be significant for interference suppression.

Declaration of competing interest

All authors disclosed no relevant relationships.

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